

PROBLEM SET 1

Show **ALL WORK** to get full/partial credit. Begin each problem on a new page, and clearly label each part of the problem. (**Max Score:** 40)

- 1) **(10 pts)** Starting from the recursion relation, obtain ψ_{210} (normalized)
hint: decompose the wave function into a radial and angular part; for the radial part, you will need to incorporate Eq. 13.1.10, but with a finite upper limit in k determined from the recursion relation. Once you have the general expression $\psi = R(r)Y(\theta, \phi)$, you may use the normalization condition $\int \psi^* \psi r^2 dr d\Omega = 1$ to determine the normalization coefficient.
- Useful integral: $\int_0^\infty x^4 e^{-2x} dx = 3/4$

- 2) **(5 pts)** The pion has a range of 1 Fermi = 10^{-5} Å as a mediator of nuclear force. Estimate its rest energy.

- 3) **(5 pts)** Estimate the de Broglie wavelength of an electron of kinetic energy 200 eV. (Recall $\lambda = 2\pi\hbar/p$.)

- 4) **(10 pts)** Instead of looking at the emission spectrum, we can also look at the absorption spectrum of hydrogen. Say some hydrogen atoms are sitting at the surface of the sun. From the interior of the sun, white light tries to come out and the atoms at the surface absorb what they can. The atoms in the ground state will now absorb the Lyman series and this will lead to dark lines if we analyze the light coming from the sun. The presence of these lines will tell us that there is hydrogen at the surface of the sun. We can also estimate the surface temperature as follows. Let T be the surface temperature. The probabilities $P(n=1)$ and $P(n=2)$ of an atom being at $n = 1$ and $n = 2$, respectively, are related by Boltzmann's formula

$$\frac{P(n = 2)}{P(n = 1)} = 4e^{-(E_2 - E_1)/kT}$$

where the factor 4 is due to the degeneracy of the $n = 2$ level. Now only atoms in $n = 2$ can produce the Balmer lines in the absorption spectrum. The relative strength of the Balmer and Lyman lines will tell us $P(n=2) / P(n = 1)$, from which we may infer T .

Show that for $T = 6000$ K, $P(n = 2) / P(n = 1)$ is negligible and that it becomes significant only for $T \approx 10^5$ K. (The Boltzmann constant is $k \approx 9 \times 10^{-5}$ eV/K. A mnemonic is $kT \approx 1/40$ eV at room temperature, $T = 300$ K .

- 5) **(5 pts)** Show that if we ignore inter-electron interactions, the energy levels of a multi-electron atom go as Z^2 . Since the Coulomb potential is Ze/r , why is the energy $\propto Z^2$?

- 6) **(5 pts)** Visible light has a wavelength of approximately $5000 \text{ \AA}$. Which of the series--Lyman, Balmer, Paschen- do you think was discovered first?
hint: determine the corresponding frequency, ω , and then check which series does the visible light frequency most closely resembles.